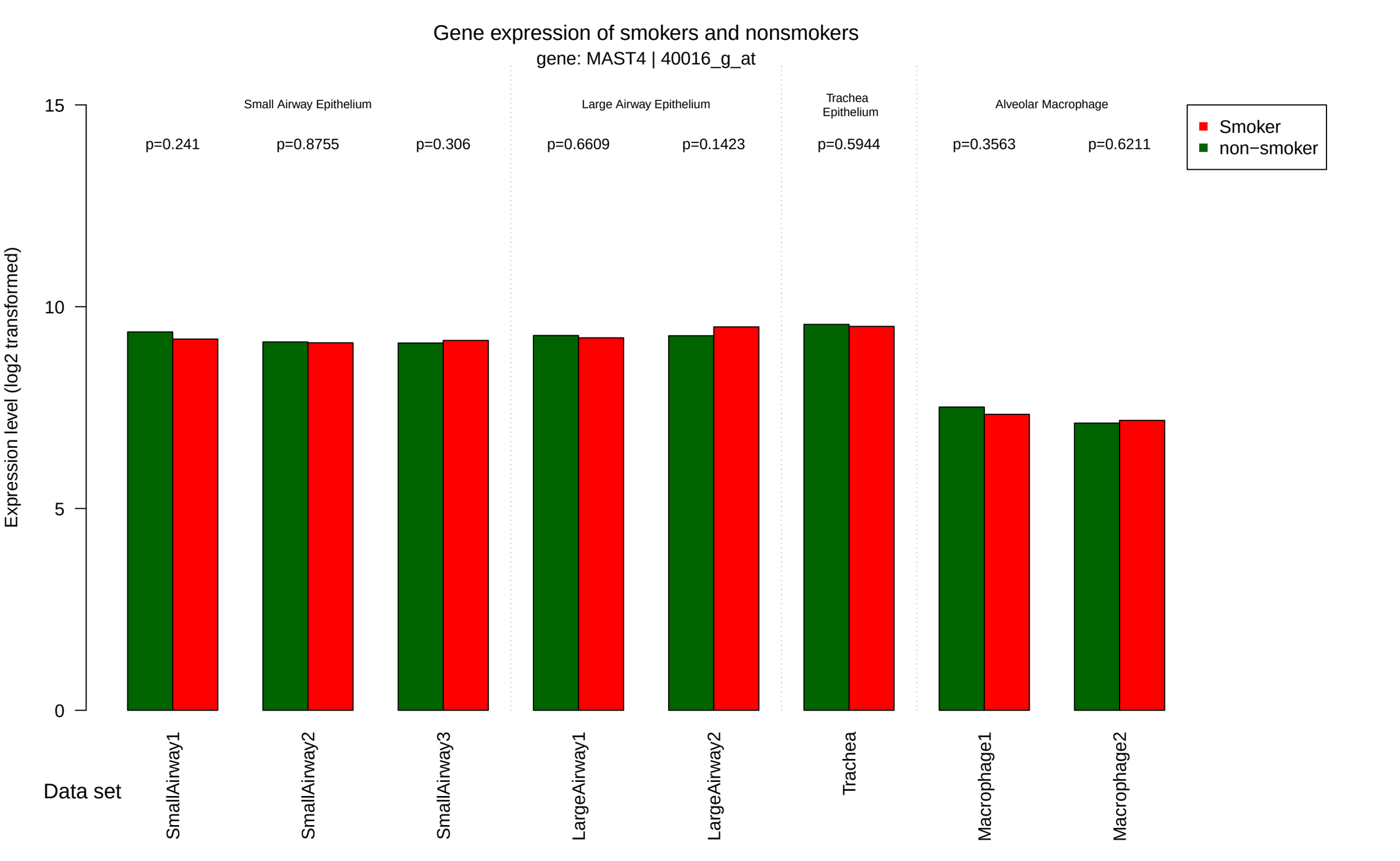
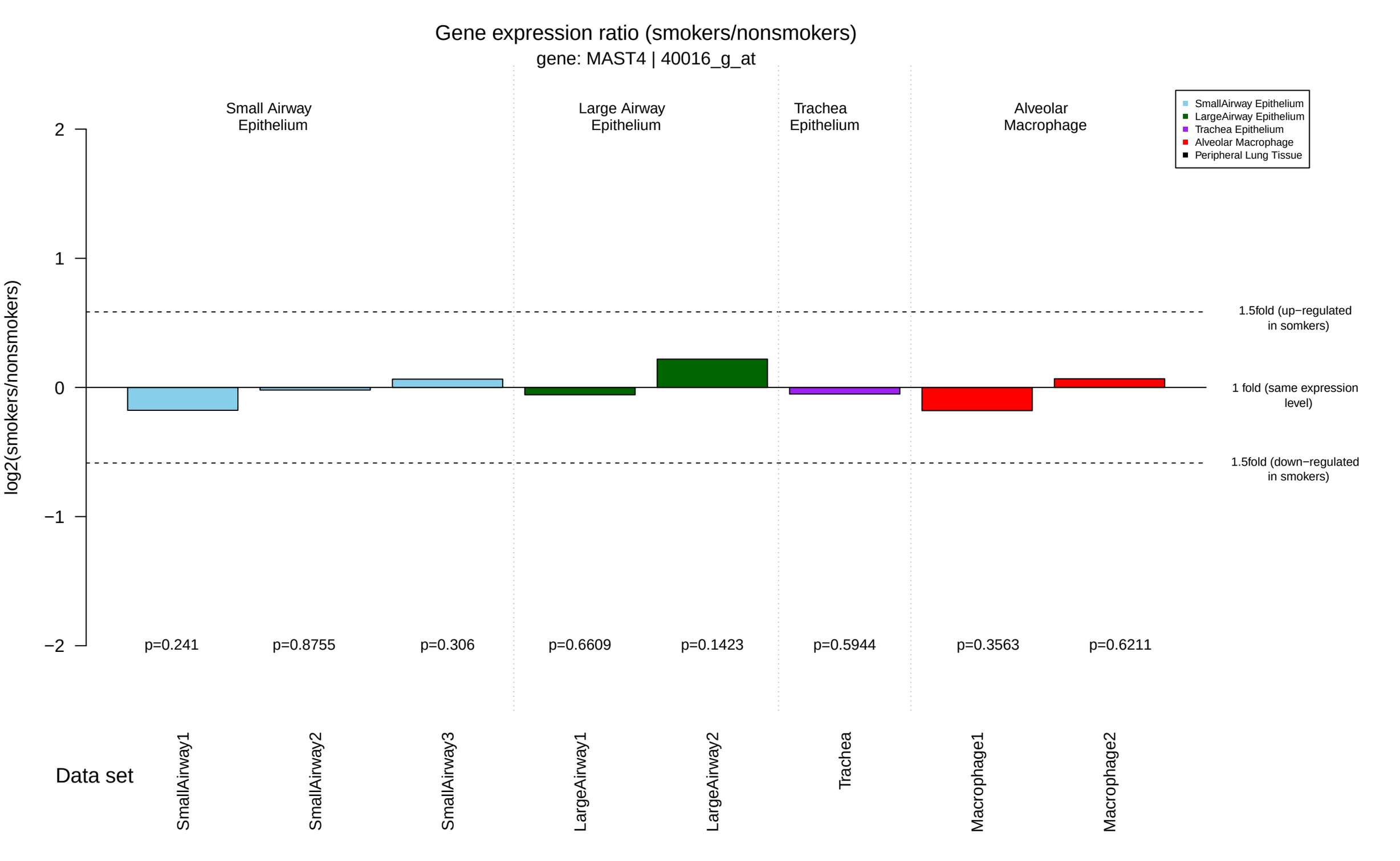
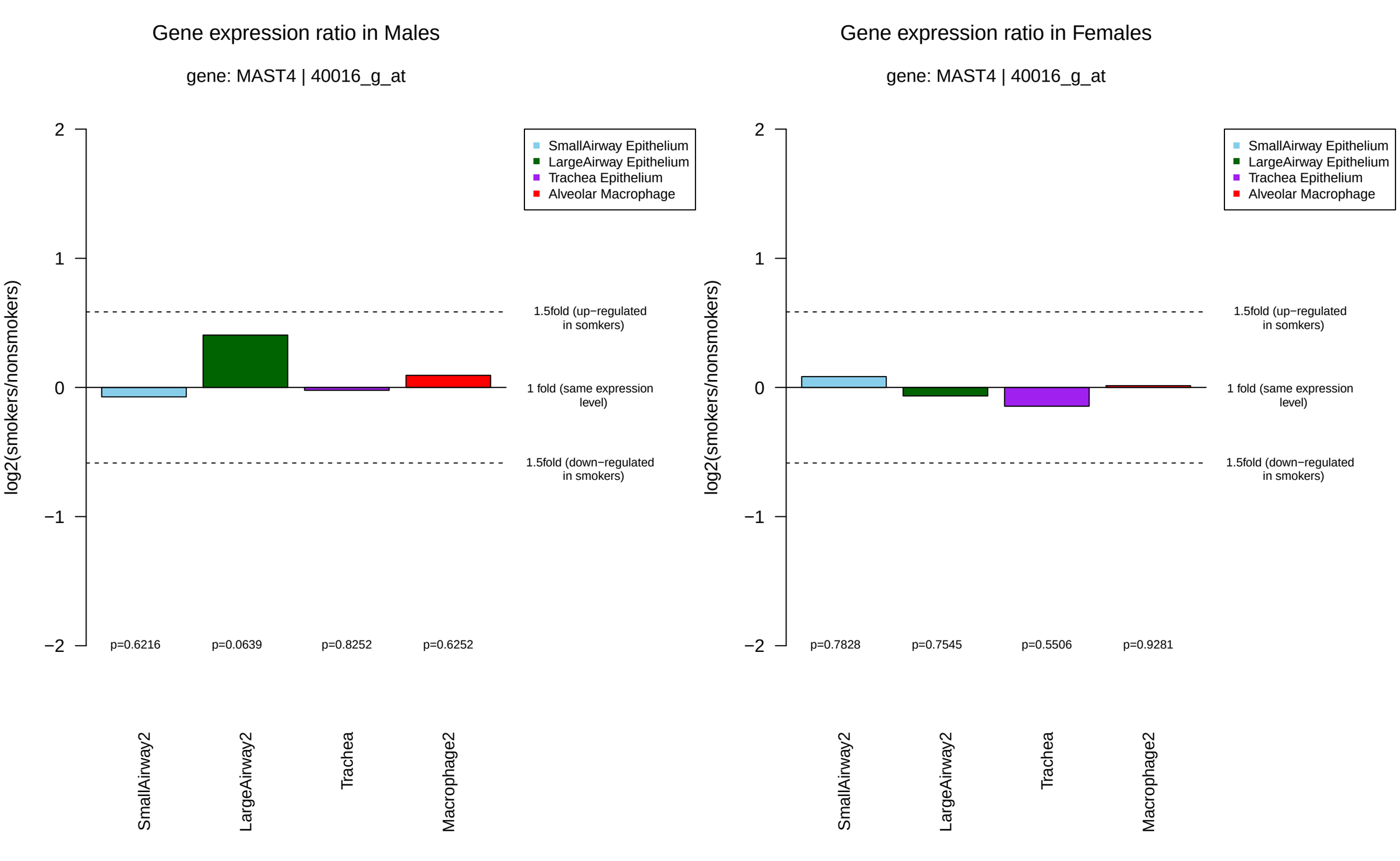
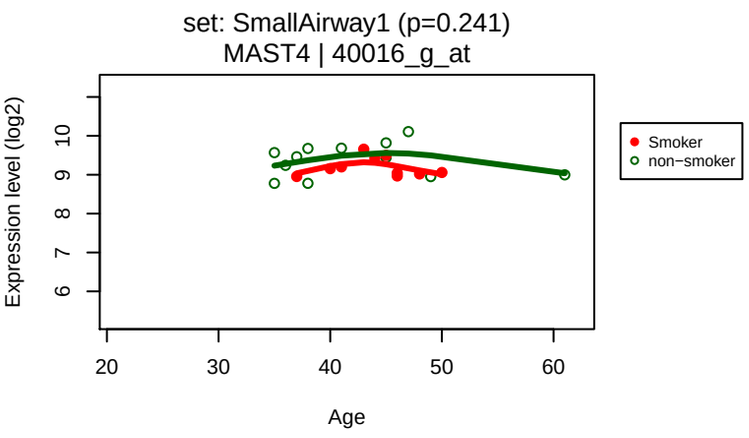
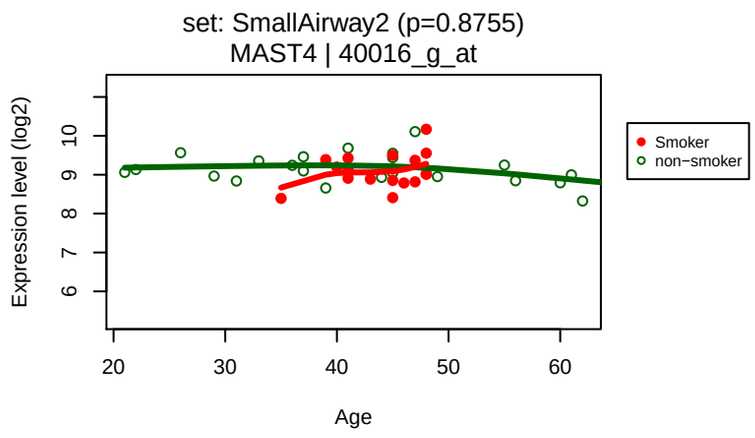
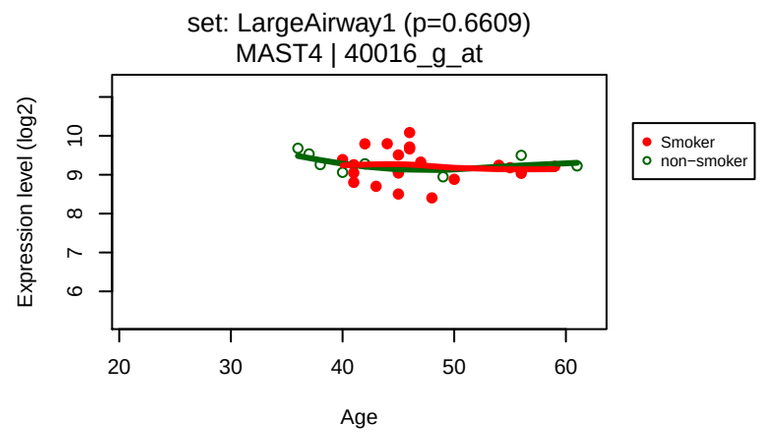

Gene expression of smokers and nonsmokers
gene: MAST4 | 40016_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.241
p=0.8755
p=0.306
p=0.6609
p=0.1423
p=0.5944
p=0.3563
p=0.6211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

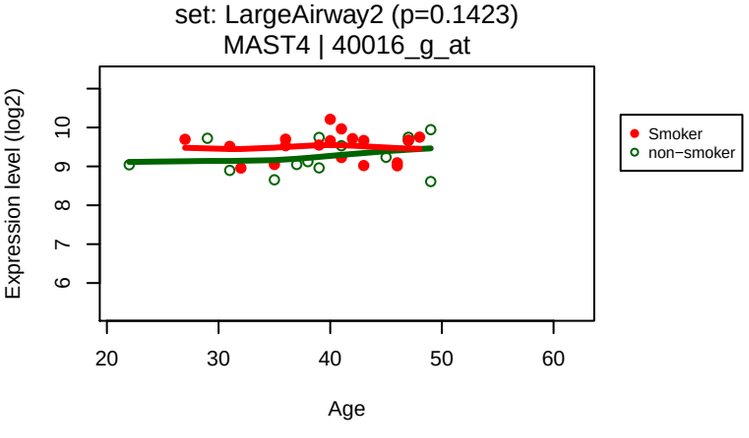

Gene expression ratio (smokers/nonsmokers)
gene: MAST4 | 40016_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.241
p=0.8755
p=0.306
p=0.6609
p=0.1423
p=0.5944
p=0.3563
p=0.6211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

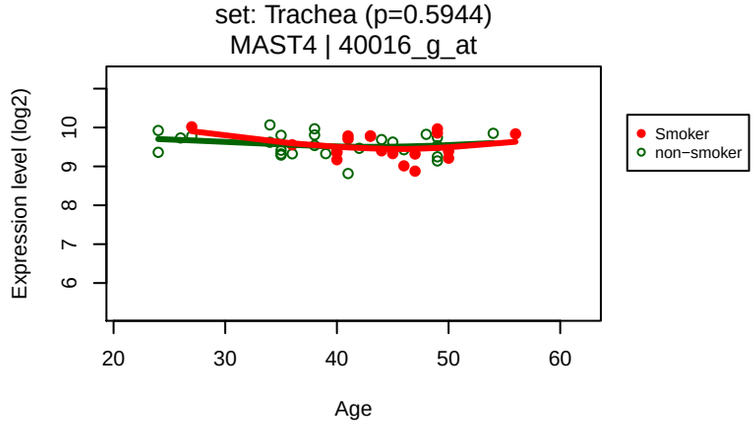

Gene expression ratio in Males
gene: MAST4 | 40016_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6216
p=0.0639
p=0.8252
p=0.6252
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAST4 | 40016_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7828
p=0.7545
p=0.5506
p=0.9281
SmallAirway2
LargeAirway2
Trachea
Macrophage2

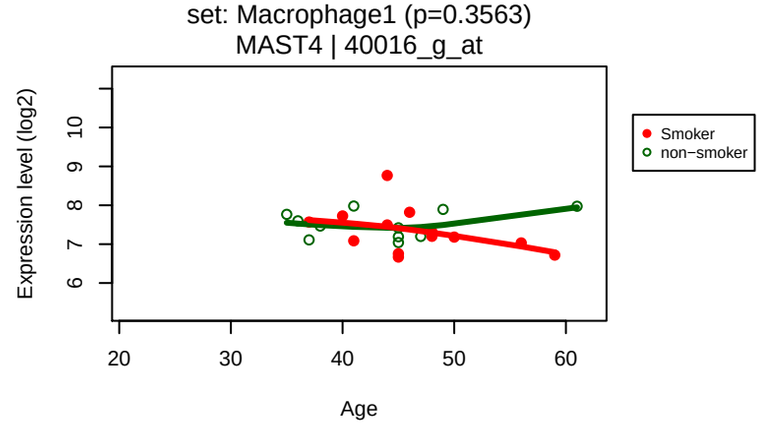

set: SmallAirway1 (p=0.241)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker

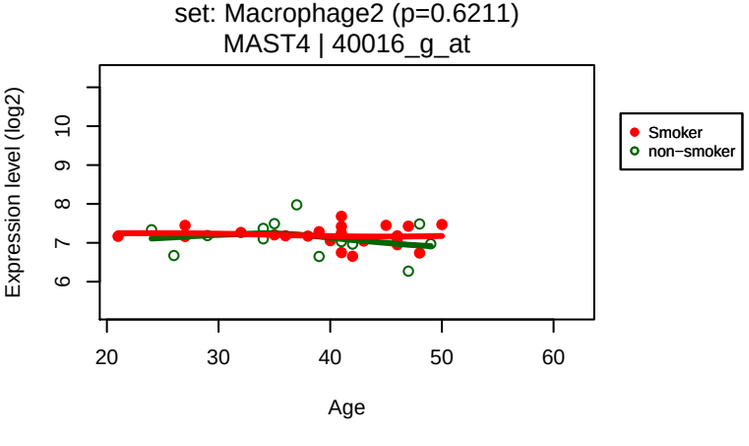

set: SmallAirway2 (p=0.8755)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker

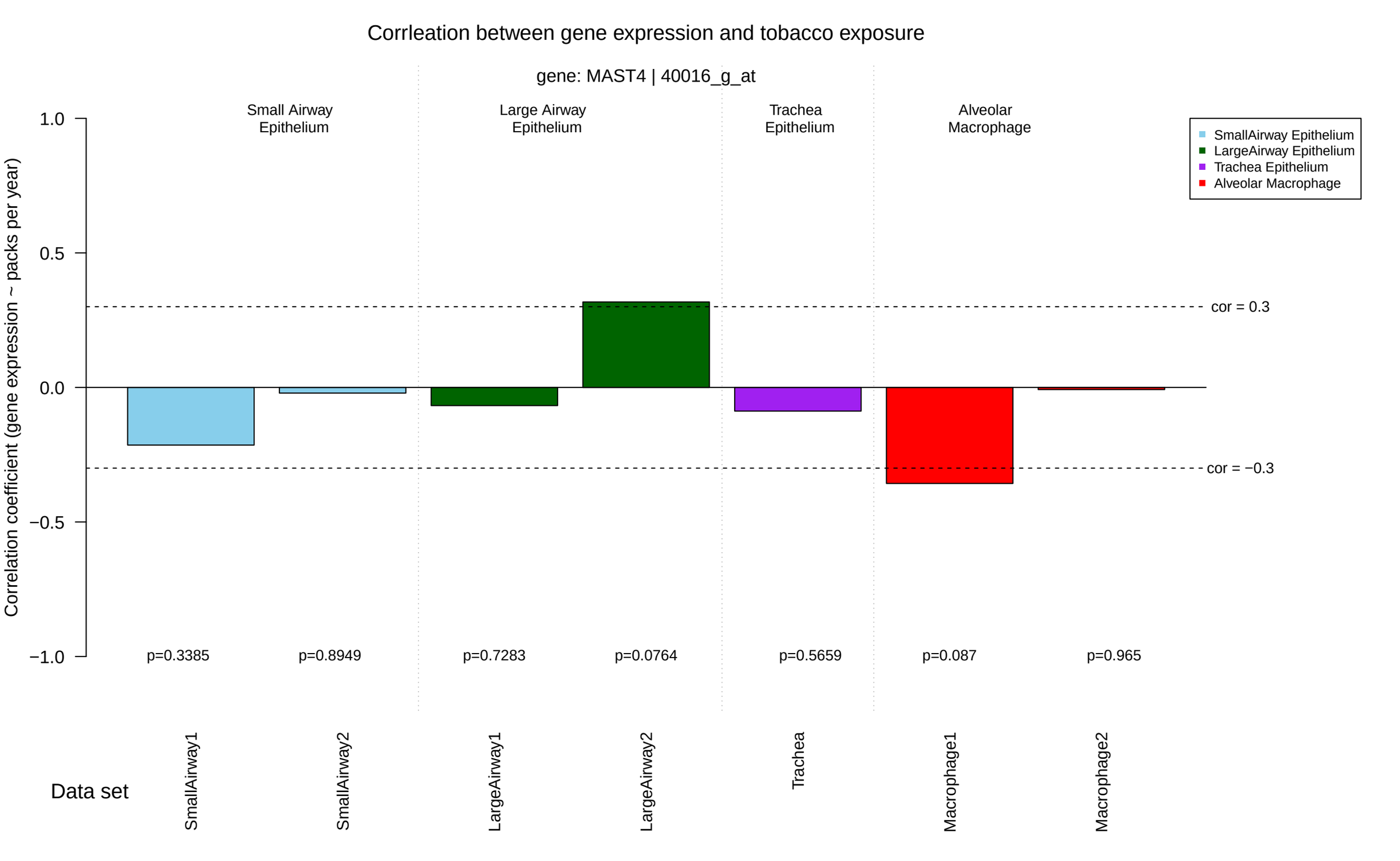

set: LargeAirway1 (p=0.6609)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1423)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5944)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3563)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6211)
MAST4 | 40016_g_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAST4 | 40016_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3385
p=0.8949
p=0.7283
p=0.0764
p=0.5659
p=0.087
p=0.965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2